

HAT-P-18b

R-2341

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_210609 · created 2026-08-19T00:25:29+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459374.81498 +/- 0.00029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.013
Transit depth (Rp/Rs)^2	3.22 +/- 0.47 [%]
Orbital Inclination (inc)	88.7 +/- 0.65
Airmass coefficient 1 (a1)	1.429 +/- 0.02
Airmass coefficient 2 (a2)	-0.327 +/- 0.057
Scatter in the residuals of the lightcurve fit is	1.4 %
Best Comparison Star	None
Optimal Aperture	4.62
Optimal Annulus	13.11
Transit Duration (day)	0.1159 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.179 $\pm$ 0.013 vs 0.1356 (deviation 3.26 σ)
Duration fit vs published	0.1159 d vs 0.1129 d (deviation 0.41 σ)
Mid-time O-C	-1.0 min
Depth SNR	13.5
Residual scatter	1.4 %

Light curve

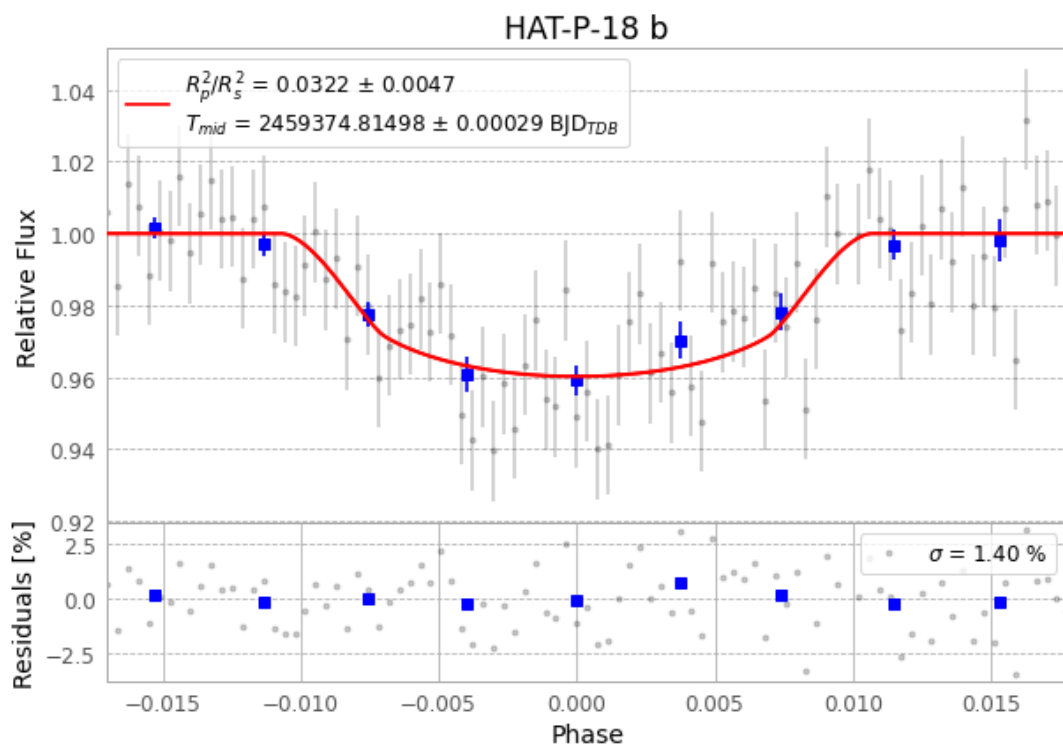


Figure 1 — FinalLightCurve_HAT-P-18 b_2021-06-08.png (SHA-256 6702288583e72ef6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [376, 186], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a252e422dd2f73b5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

93 FITS frames (61 MB), HATP-18210609051212.FITS → HATP-18210609094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-210609.log (4b6a933a85ca...), FinalLightCurve_HAT-P-18 b_2021-06-08.png (6702288583e7...), FinalLightCurve_HAT-P-18 b_2021-06-08.csv (eec66db499b4...)

Compute resources

Wall time 200.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 00:25Z.